



ORIGINAL RESEARCH ARTICLE

Application of the Small Punch Creep Test in Predicting Creep Properties of Grade 91 Steel

Imam Ul Ferdous, Nasrul Azuan Alang, Juliawati Alias, and Norhaida Ab. Razak

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This study evaluates the effectiveness of small punch creep testing (SPCT) as an alternative to conventional uniaxial creep testing (UCT) in predicting creep properties for Grade 91 steel. Cylindrical specimens extracted longitudinally from an as-received Grade 91 pipe were subjected to UCT at 600 °C under the stress level of 160–190 MPa. Equivalent SPCTs utilized a custom-designed rig to test 0.5-mm-thick, 8-mm-diameter disc specimen at 600 °C under the load level of 400–475 N loads. Both test types exhibited typical primary, secondary, and tertiary creep regimes. A correlation factor of 2.5 provided excellent agreement between SPCT and UCT rupture times. Converting SPCT punch displacement rates to equivalent minimum creep strain rates enabled application of established creep models like Monkman–Grant and Norton for long-term rupture life prediction. The Larson–Miller approach successfully correlated short-term SPCT data for long-term rupture life extrapolation. Minimum creep strain rate versus stress trends from UCT aligned well with equivalent rates from SPCT data. Fractographic analysis revealed similar ductile, dimple-dominated fracture morphologies in both test types. This study validates SPCT as an efficient methodology for creep assessment using minimal material volumes.

Keywords correlation, creep, Grade 91 steel, rupture, small punch

1. Introduction

The swift evolution of modern civilization and technological progress has significantly escalated the demand for energy. Consequently, power plants must ensure maximum output to meet the increasing demand, equating to maximum efficiency. The operating temperature of these plants directly influences their efficiency and environmental impact, with higher temperatures generally leading to improved performance and reduced emissions (Ref 1). However, this adjustment exposes plant components to the phenomenon of creep. Creep, the slow, time-dependent deformation of materials under constant stress, is a critical concern for components exposed to high temperatures over extended periods. This can lead to significant changes in material properties, ultimately affecting the performance and safety of the component (Ref 2). While uniaxial creep testing (UCT) has been the standard for studying creep behavior, it requires substantial material and time investments, sometimes 400 to 4000 h (Ref 3). The need for more efficient testing methodologies has led to the exploration of alternative approaches, such as the small punch creep testing (SPCT) technique (Ref 4). Over the past few years, the small punch creep test technique has attracted growing research interest and

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List of symbols

$\dot{\epsilon}_{\min}$	Minimum creep strain rate
C_{MG}	Material constant in Monkman–Grant model
T_0	Initial thicknesses
d_0	Height of the moving contact boundary
t_r/t_f	Rupture time/rupture life
u_0	The angle at the moving contact boundary
ϵ_{eq}	Equivalent strain
ϵ_r	Radial strain
Ψ	Correlation factor
A	Temperature-dependent material constant in Norton creep model
C	Material-specific constant
M	Temperature-dependent material constant
R	Radius of the punch
T	Current thickness
a	Radius of the disc between supports
d/h	Punch displacement/central deflection of the disc
m	Creep exponent in Monkman–Grant model
n	Creep exponent in Norton creep model
r	Current radius
u	The angle at the disc's edge
v	Creep exponent
z	Height of an element
σ	Applied stress
ω	The angle made by the surface normal with the vertical axis

Abbreviations

EDM	Electric discharge machining
LMP	Larson–Miller parameter
LVDT	Linear variable differential transformer
MCSR	Minimum creep strain rate
SEM	Scanning electron microscope

SPCT	Small punch creep test
SPT	Small punch test
UCT	Uniaxial creep test

been increasingly embraced by scientists across the globe (Ref 5-10). SPCT represents a paradigm shift in the assessment of creep properties, offering a path to rapid, material-efficient evaluation. This method's ability to derive meaningful creep behavior data from small, non-standard specimens presents a significant advantage, particularly for in-service components where material availability is restricted. The integration of the SPT method into the realm of material testing has brought about a transformative approach, especially in the evaluation of creep properties of materials. This technique, which involves the use of minuscule disc-shaped specimens with diameters ranging from 8 to 12 mm and a thickness of merely 0.5 mm, presents a less invasive means of assessing the integrity of components currently in service (Ref 11). The minimal material requirement and the expeditious nature of the test offer a stark contrast to the traditional, more cumbersome creep tests that require larger samples and potentially compromise the component under examination. Despite the numerous advantages that the SPT method offers, including its application in a variety of contexts from nuclear to petrochemical industries, the translation of its results to practical engineering designs is not straightforward. This is primarily due to the distinct stress conditions encountered in SPT compared to those in conventional creep tests. The challenge lies in establishing a robust correlation between the outcomes of the SPT such as the SP load and the central deflection and standard creep test parameters like uniaxial creep stress and strain (Ref 12). Efforts to delineate this relationship have led to the development of various methodologies, some of which have been encapsulated in standards such as the CWA-15627 (Ref 13). These methods frequently present a correlation coefficient between the small punch load and the uniaxial creep stress, even though this coefficient is recognized to vary with changes in material kinds and testing circumstances, underscoring the intricacy and necessity for a more refined tactic to precisely determining it. Originally conceived for assessing the transition from brittle to ductile behavior and later adapted for other material property evaluations such as fracture toughness and material degradation, the SPT technique has broadened its utility. As an efficient alternative to the UCT, the SPCT method conserves resources by utilizing significantly smaller specimens and reduces the time required to obtain test results. However, the interpretation of SPCT-derived data, particularly in terms of displacement and load, into meaningful strain and stress values pertinent to UCT, remains a complex endeavor due to the multi-axial stress state of SPCT specimens. The advent of Grade 91 steel, also known as 9Cr-1Mo, marked a significant milestone in the development of materials for high-temperature applications. Originating from the quest for superior efficiency and reliability in power generation systems, Grade 91 steel embodies a remarkable blend of strength, durability, and resistance to thermal degradation, making it an indispensable

asset in the energy sector (Ref 14). Its unique composition, characterized by a martensitic structure enhanced with chromium, molybdenum, and vanadium, offers exceptional corrosive resistive performance under the rigorous conditions of high pressure and temperature prevalent in modern supercritical and ultra-supercritical power plants (Ref 15). The critical role of Grade 91 steel extends beyond the energy sector, finding applications in the petrochemical industry, where materials are subjected to harsh environments, and in the aerospace sector, where reliability and performance are paramount. The material's ability to maintain structural integrity and resist creep deformation under prolonged exposure to high temperatures has been a key factor in its widespread adoption. However, the increasing demands on these materials, driven by the push toward higher efficiencies and longer lifespans, necessitate a deep understanding of their long-term behavior under stress.

This study presents several novel contributions to the field of creep behavior analysis for Grade 91 steel, significantly advancing both theoretical understanding and practical applications. Uniquely, it offers the comprehensive, side-by-side comparison of small punch creep testing (SPCT) using newly developed test rig and uniaxial creep testing (UCT) results across an unprecedentedly wide range of stresses, providing unparalleled insights into the material's behavior under diverse conditions. The research introduces a methodology for refining the correlation between SPCT and UCT data, employing advanced statistical techniques to enhance the applicability of SPCT in real-world scenarios. This study integrated multiple creep models (Monkman-Grant, Norton, and Larson-Miller) with both SPCT and UCT data, offering a more holistic approach to creep behavior prediction. Furthermore, it combines detailed fractographic analysis with creep data from both testing methods, providing a deeper understanding of the microstructural changes and failure modes associated with creep in Grade 91 steel.

2. Mathematical Formulation

2.1 Deformation under Small Punch Load

Numerous theoretical frameworks to analyze the data obtained from small punch creep tests have been reported. Of the various models proposed, the one formulated by Chakrabarty (Ref 16) involving membrane stretching has gained widespread acceptance and application. The mechanics of deformation of the disc under small punch load, according to Chakrabarty, is shown in Fig. 1. The model starts with the assumption that the relationship between central deflection (d) and central strain is geometric in nature. This is based on stretch forming over a hemispherical punch head, where the material is considered to be rigid, and the deformation occurs under membrane stresses only due to the small thickness of the disc sample compared to the punch radius.

The disc during deformation forms a surface of revolution with its axis coinciding with the vertical axis of the punch. a is the radius of the disc between supports, R is the radius of the punch, r and z represent the current radius and height of an element, respectively, and ω is the angle made by the surface normal with the vertical axis. u is the angle at the disc's edge ($r = a$), and u_0 is at the moving contact boundary. d is the

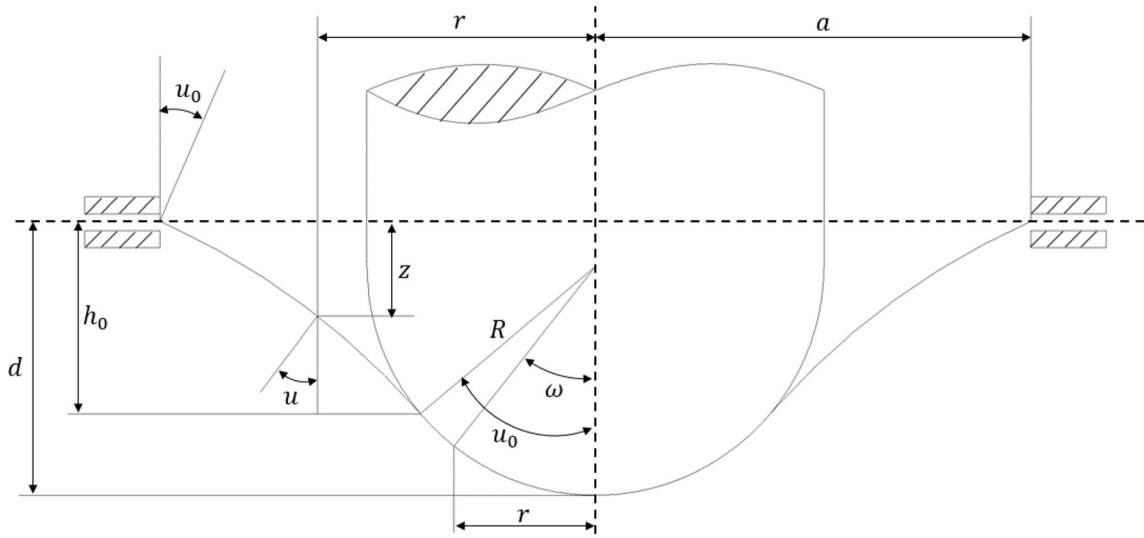


Fig. 1 Geometry of the Chakrabarty membrane stretch model

central deflection of the disc, and d_0 is the height of the moving contact boundary. Chakrabarty derived the expression for the von Mises equivalent strain in both the contact region

($0 \leq \omega \leq u_0$) and the non-contact region ($u \leq \omega \leq u_0$), obtaining the maximum equivalent strain at the center of the disc ($\omega = 0$) as:

$$\epsilon_{eq} = \ln \left[\frac{2 + 2 \cos u}{(1 + \cos u_0)^2} \right] + \ln \left[\left(\frac{1 + \cos u}{1 + \cos u_0} \right) \right] \quad (\text{Eq 1})$$

The central deflection of the disc, which is equal to the value of z when $\omega = 0$, is given by:

$$d = R \sin u_0 \ln \left(\frac{\tan \frac{u_0}{2}}{\tan \frac{u}{2}} \right) + R(1 - \cos u_0) \quad (\text{Eq 2})$$

A relation connecting u and u_0 is established as:

$$\sin u = \frac{R}{a} \sin^2 u_0 \quad (\text{Eq 3})$$

The angle u_0 , which represents the angle at the moving contact boundary, is varied within the range from 0° to 90° . This range covers the possible angles that could be formed by the surface normal at the moving contact boundary with the vertical axis, considering the entire possible deformation of the disc from no deformation (0°) to full deformation (90°). For each value of u_0 within the specified range, the corresponding values of central deflection (d) and maximum equivalent strain (ϵ_{eq}) are computed using Eqs 1, 2, and 3 derived by Chakrabarty. This results in a set of d and ϵ_{eq} values that reflect the relationship between the central deflection and strain for the given specimen sizes and range of u_0 . The computed sets of d and ϵ_{eq} values are then fitted using a polynomial expression. This fitting process involves finding a polynomial equation that best describes the relationship between d and ϵ_{eq} based on the computed data points. The polynomial equation represents a mathematical model that can predict the maximum equivalent strain based on a given central deflection within the context of the specified specimen sizes and deformation conditions. The outcome of the polynomial fitting is a relationship between central deflection (d , in units of mm) and maximum equivalent strain (ϵ_{eq}). Employing $a = 2.20$ mm

and $R = 1.25$ mm, this relationship is expressed as a polynomial equation, which, in this context, is given as (Ref 17):

$$\epsilon_{eq} = 0.233818d^2 + 0.40616d \quad (\text{Eq 4})$$

This sequence of derivations connects the geometric parameters and material properties with the central deflection, ultimately relating the central deflection to the maximum equivalent strain through a polynomial equation.

Lee et al. (Ref 18) introduced another novel approach for determining equivalent strain, which facilitates the translation of displacement measurements from SPCT into strain metrics relevant to UCT. This methodology is underpinned by the theoretical framework of Chakrabarty's membrane stretching principles (Ref 16). Within this context, the equivalent strain is expressed as:

$$\epsilon_{eq} = |\epsilon_r| = \ln \left(\frac{T_0}{T} \right) \quad (\text{Eq 5})$$

In the context of the study, ϵ_r represents the radial strain oriented along the specimen's thickness axis. The variables T and T_0 denote the current and initial thicknesses of the specimen, respectively. Progressing further, the research incorporates Mao's empirical formula (Ref 19) to delineate the specimen's thickness at a given punch displacement $T(h)$, which is expressed as:

$$T(h) = T_0 e^{-1.2 \left(\frac{h}{2} \right)^{1.5}} \quad (\text{Eq 6})$$

where h signifies the displacement induced by the punch. Upon integrating the established Eq 6, which specifies the specimen's thickness corresponding to a particular punch displacement, into Eq 5, a refined expression for the equivalent strain is subsequently derived as follows:

$$\epsilon_{eq} = 1.2 \left(\frac{h}{2} \right)^{1.5} \quad (\text{Eq 7})$$

Leveraging Eq 7, the formulation for equivalent strain proposed by Lee et al. (Ref 18) enables the acquisition of strain metrics pertinent to UCT, essential for the assessment of creep characteristics.

2.2 Models for Assessing Creep Properties

2.2.1 Monkman–Grant Model. The Monkman–Grant formulation (Ref 20) has emerged as a seminal approach for predicting the creep life of materials based on creep rate parameter. This formulation is characterized by its elegant simplicity, yet it offers profound insights into the creep behavior of materials. The Monkman–Grant relation postulates that the minimum creep rate ($\dot{\epsilon}_{\min}$), which represents the steady-state rate of creep deformation, is inversely proportional to the time to fracture (t_f), the total duration a material can withstand creep conditions before failure. Mathematically, this relationship can be expressed as:

$$\dot{\epsilon}_{\min}^m t_f = C_{MG} \quad (\text{Eq 8})$$

In this equation, m and C_{MG} is a constant specific to a given material under particular conditions of temperature and stress. While this constant varies with the material and environmental conditions, it remains invariant for a wide range of stress levels and temperatures within a given material system. Typically, the material constant m ranges from 0.8 to 0.95, while the constant C_{MG} falls within the range of 3 to 20 (Ref 21). The significance of the Monkman–Grant formulation lies in its predictive capability. By determining the minimum creep rate through short-term tests, it becomes feasible to predict the long-term creep life (t_f) of a material under similar temperature conditions.

2.2.2 Norton Creep Model. The Norton's power law (Ref 22) stands as a fundamental model that elucidates the steady-state creep behavior of materials. This empirical relationship establishes a direct correlation between the minimum creep rate ($\dot{\epsilon}_{\min}$), and the applied stress (σ).

The mathematical formulation of the Norton creep law is expressed as:

$$\dot{\epsilon}_{\min} = A\sigma^n \quad (\text{Eq 9})$$

In this equation, A denotes the creep coefficient, a material-dependent constant, σ represents the applied stress, n is the stress exponent, a material-dependent constant typically ranging from 3 to 8. The Norton creep law elucidates the creep strain rate exhibits a nonlinear relationship with the applied stress, governed by the stress exponent (n), where a higher value of ' n ' signifies a greater sensitivity of the creep rate to changes in stress.

2.2.3 Larson–Miller Parameter. The Larson–Miller parameter (LMP) has emerged as a valuable tool enabling the prediction of the time to rupture (failure) of materials under creep conditions. This parameter builds upon the time-temperature equivalence observed in creep behavior, recognizing that higher temperatures accelerate creep processes, leading to potential failure in a shorter time compared to lower temperatures under the same stress conditions. The LMP provides a means to consolidate creep data acquired at various temperatures and stresses into a single master curve. The formulation of the Larson–Miller parameter is as follows:

$$LMP = T(C + \log t_r) \quad (\text{Eq 10})$$

where T represents the absolute temperature (in Kelvin), t_r denotes the time to rupture (in hours), and C is a material-specific constant.

3. Materials and Methods

3.1 Chemical Composition of Grade 91 Steel

The chemical composition of the Grade 91 steel used in this study was determined using optical emission spectroscopy (OES) on a polished cylindrical sample extracted from the as-received pipe block. The average values of five distinct analysis points are presented in Table 1.

3.2 Microstructure of As-Received Grade 91 Steel

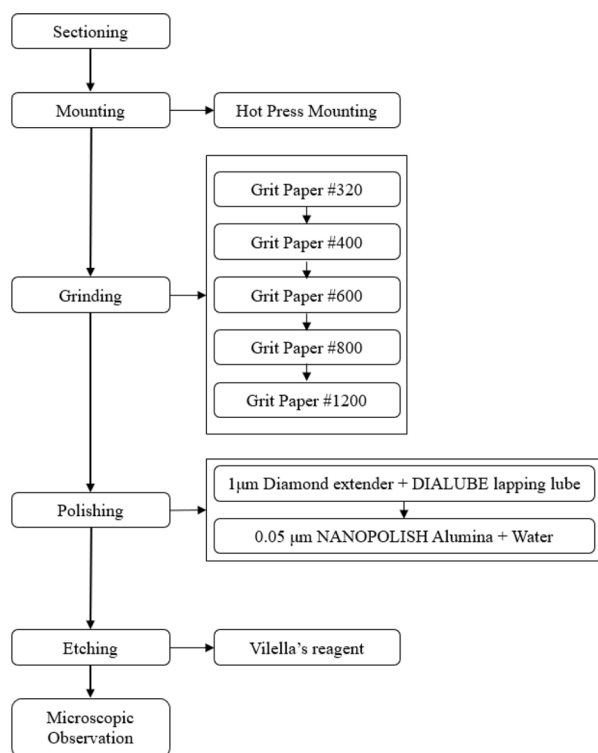
A cylindrical specimen with a diameter of 10 mm and a height of 10 mm was extracted from a Grade 91 pipe in its as-received condition for microscopic examination. The sample was mounted using a hot mounting machine, wherein a thermoset plastic material of Bakelite was employed to embed the specimen. The mounted specimen underwent a grinding process, involving sequential abrasion with sandpapers of incrementally finer grits (#320, #400, #600, #800, and #1200). After achieving uniform grinding in one direction, the specimen was rotated 90° and subjected to the subsequent finer grit sandpaper. This procedure was repeated until the desired surface finish was attained. Subsequent to the grinding stage, the specimen was polished in two distinct steps. The initial polishing phase involved the use of a 1 μm diamond extender and DIALUBE lapping lubricant, followed by a second polishing step employing 0.05 μm NANOPOLISH alumina and water, yielding a mirror-like surface finish. During the polishing process, the specimen was held against the direction of the rotating polishing cloth to mitigate the effects of directional polishing on the metal surface. After polishing, the specimen was carefully cleaned to remove residual grease, oil, and polishing compounds. Thereafter, it was etched with Vilella's reagent for 10-15 seconds, followed by rinsing under running water and drying with an air gun. The etched sample was then examined under an optical microscope and scanning electron microscopy (SEM), enabling the capture and analysis of the revealed microstructural features. Figure 2 shows the process flow of the specimen preparation for microstructural analysis.

3.3 Uniaxial Creep Testing

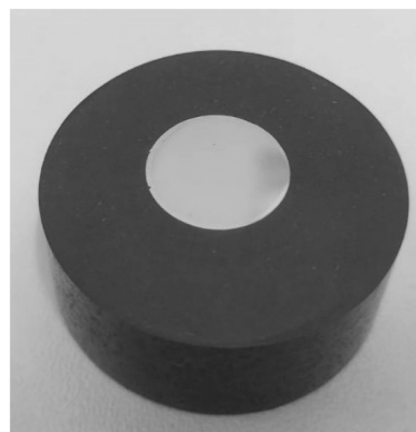
All specimens for the UCT were taken longitudinally from a Grade 91 pipe that was commercially available in its as-received condition (see Fig. 3a). As depicted in Fig. 3(c), the UCT specimen bars had a total length of 92 mm, a gauge length of 36 mm, and a gauge diameter of 8 mm. Creep experiments were carried out at 600 °C, applying stress (σ) levels from 160 to 190 MPa in increments of 10 MPa, as outlined in the overview provided in Table 2. The testing procedures adhered to the established guidelines of ASTM E139 (Ref 23). A dead-load creep testing machine furnished with a cylindrical furnace (see Fig. 4) enabled the tests. The threaded specimen holder, lubricated with Loctite silver-grade anti-seize lubricant for high-temperature protection, secured the specimen within the furnace. To measure longitudinal strain, two gripping mechanisms connected the specimen to linear variable differential transformers (LVDTs) positioned at the furnace bottom (see Fig. 4). Furthermore, three K-type thermocouples connected to the specimen at key locations (top, middle, and bottom) checked the temperature with a maximum variation of 2 °C during the entire test. The specimen was gradually heated to

Table 1 Chemical composition of Grade 91 steel

C	Si	Mn	P	S	Cr
0.108	0.410	0.380	0.003	0.005	8.790
Mo	V	Nb	N	Ni	Al
0.994	0.226	0.040	0.045	0.100	0.005



(a)



(b)

Fig. 2 (a) Process flow of the sample preparation for microscopic observation and (b) a mirror-like surface of the sample

600 °C, ensuring consistent temperature distribution across its surface. Upon reaching the target temperature, the pre-defined load was applied. The LVDTs recorded the specimen's axial displacement throughout the test, which continued until rupture. The LVDTs recorded unprocessed data as axial movement throughout the experiment. This information will later be examined to comprehend the creep characteristics of Grade 91 steel under the given conditions.

3.4 Small Punch Creep Testing

The small punch creep test specimen employed in this study follows to the latest code of practice, possessing a diameter of 8 mm and a thickness of 0.5 mm (Ref 24). To fabricate the specimen, a longitudinal section of bar shaped with 8 mm diameter was extracted from the as-received Grade 91 pipe. Then, electric discharge machining process was utilized to create a disc-shaped specimen with a thickness of $h + 0.1$ mm. Further refinement was achieved through bilateral grinding using abrasive paper of varying grit sizes, ensuring removal of at least 0.03 mm of material from each side for optimal

uniformity. Given the small size of the specimen, a custom-designed holder was developed for secure and stable grinding and polishing. The specimen's thickness was meticulously measured at four equidistant points around the perimeter at 90° intervals, along with a central measurement, as detailed in Fig. 5. Each measurement was verified to be within the specified tolerance limit of ± 0.005 mm. While the diameter is less critical due to clamping during testing, it was also confirmed to conform to the tolerance established in the code of practice. Although less critical, the diameter was also confirmed to conform to the established tolerance.

To leverage the existing uniaxial creep testing machine for small punch creep tests, a custom-designed rig was conceptualized and fabricated. This conversion necessitated a shift from tensile to compressive loading. The final design (illustrated in Fig. 6) features a stainless-steel grade 310 construction, chosen for its excellent machinability and high-temperature endurance. Two threaded studs facilitate secure integration with the existing machine. Two rectangular flat bars, attached to the moving bottom section of the rig, extend toward the LVDT to capture punch displacement. These bars incorporate guides for

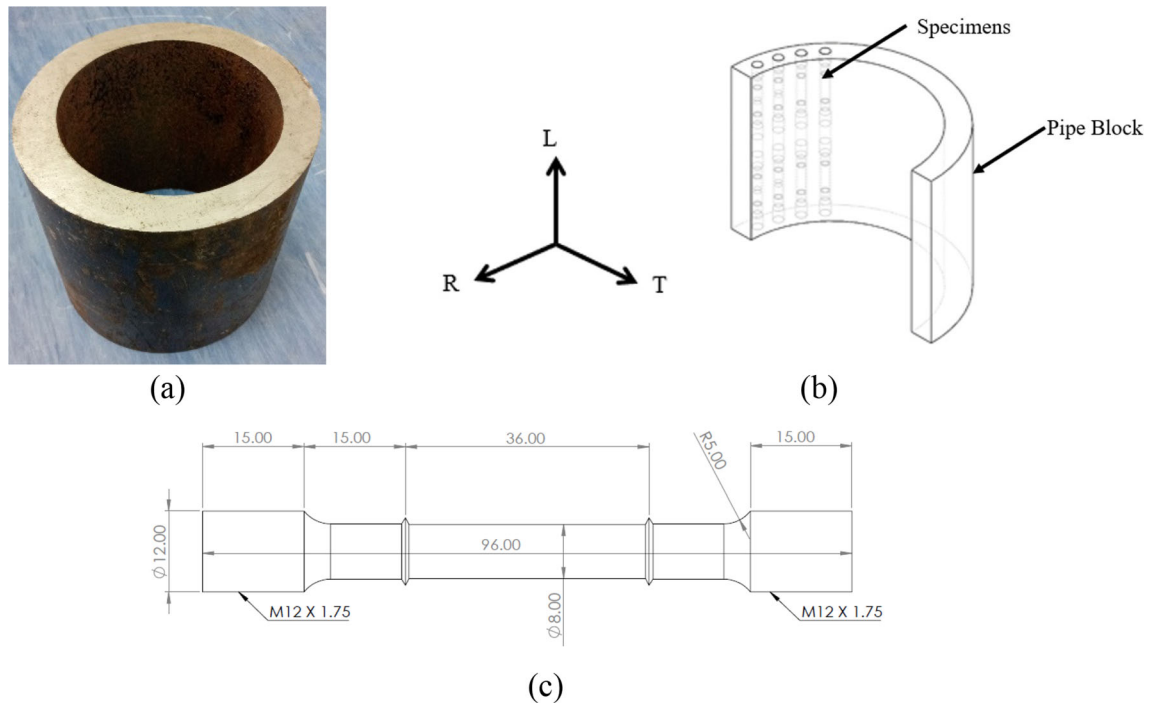


Fig. 3 (a) Block of P91 pipe, (b) illustrative diagram showing the removal of the bar, (c) specifics of the specimen's shape and dimensions

Table 2 Experimental plan for the UCT

ID-specimen	Type of the specimen	σ , MPa	T , °C
UCT_160	As-received specimen	160	600 °C
UCT_170		170	
UCT_180		180	
UCT_190		190	

smooth movement. The upper and lower dies of the specimen holder are positioned within the rig. The punch and its corresponding hole are precisely engineered to limit punch movement beyond a predetermined threshold, ensuring controlled and accurate testing conditions.

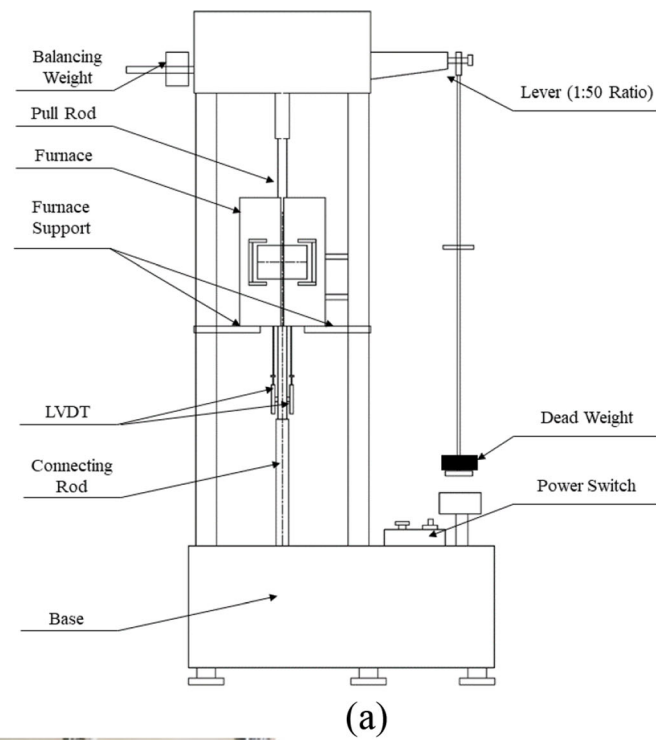
The disc-shaped specimen was securely clamped between the upper and lower dies. A flat-faced punch equipped with a 2.5-mm ceramic ball applied the compressive force. Note that no lubricant is applied between the contacting surface between the specimen and puncher. While the friction may affect the specimen deformation behavior, however, this parameter is kept constant throughout the test program. Prior to testing, both the creep testing machine and LVDT underwent calibration. Tests were conducted in an ambient atmosphere, assuming negligible impact of oxidation on the results. The tubular furnace surrounding the test rig ensured precise temperature control. Three K-type thermocouples were carefully placed at the upper, middle, and lower parts of the test rig close to the sample (illustrated in Fig. 6), offering ongoing temperature surveillance. Throughout the test, the maximum deviation between thermocouple readings was maintained within 2 °C. The test rig, secured with two threaded studs, housed the testing rig within the furnace. To prevent high-temperature corrosion and seizing, Loctite silver-grade anti-seize lubricant was applied to the stud threads and other crucial components before insertion.

Once the target temperature of 600 °C was attained across all thermocouples, a pre-defined load was applied. Figure 6 details the experimental setup, while Table 3 summarizes the test matrix.

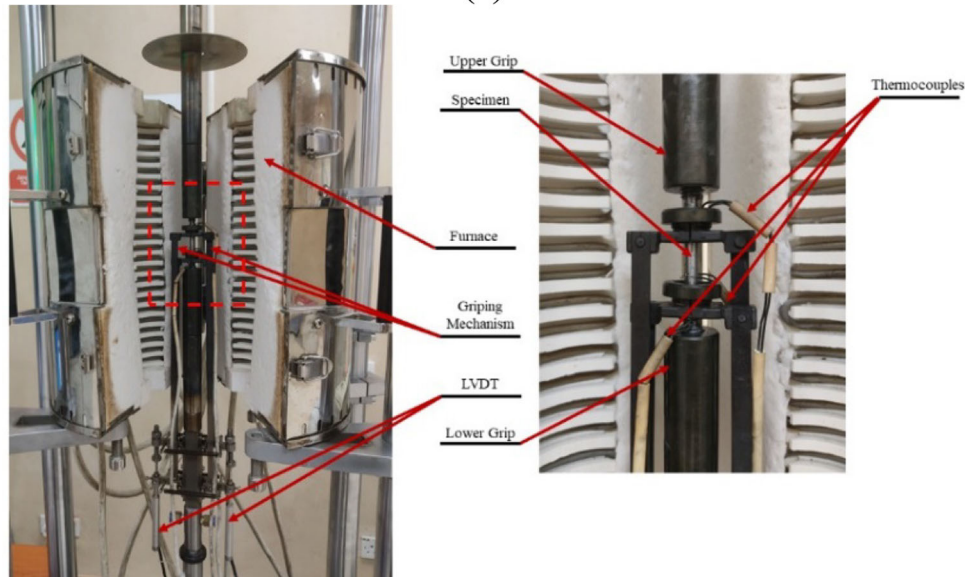
4. Results and Discussion

4.1 Creep Curve

Figure 7 exhibits the relation between creep strain and time for uniaxial creep test subjected to stress magnitudes spanning from 160 to 190 MPa. These tests demonstrate a typical three-stage deformation process: primary, secondary, and tertiary. Initially, in the primary phase, there is a high strain rate under the applied load, which gradually diminishes. The material experiences a deformation ranging from 0.025 to 0.05 mm/mm prior to entering the secondary stage, at which point a consistent creep strain rate is observed. This stage is marked by a balance in the formation and disintegration of high dislocation density within the material, as identified by Sakthivel et al. (Ref 25). Concurrently, the material experiences ongoing creep damage, diminishing its creep strength and precipitating the tertiary stage. In this concluding phase, the material exhibits tensile instability or necking, which results in a significant acceleration of the strain rate up to the point of fracture. Complementing these findings, Fig. 8 elucidates the correlation between punch displacement and loading time under various loads (400N-475N) at 600 °C, using data from SPCTs. The time-dependent deflection patterns of the same steel (Grade 91) closely resemble those observed in standard UCTs, exhibiting similar three-phase deformation. This includes an initial phase with a decreasing deflection rate, followed by a steady-state phase with a consistent rate, and concluding with an accelerating phase culminating in rupture.



(a)



(b)

Fig. 4 (a) Creep testing machine, (b) experimental setup

The constant deflection rate noted in the middle phase is akin to the least creep rate seen in UCT. Table 4 compiles the rupture times for both test types under the specified temperatures and stresses/loads, providing key data for subsequent analysis. The small punch creep tests successfully replicated the three creep regimes displayed in uniaxial testing, enabling direct comparison between the two methodologies.

4.2 Determination of Correlation Factor (W)

This section focuses on establishing a correlation factor linking the results of SPCT to UCT through detailed comparative analysis. As outlined in the BSEN 10371:2021 standard (Ref 24), careful determination of the applied load in SPC

testing is crucial. This determination depends on both geometric factors like specimen thickness and intrinsic material properties. The goal is to ensure the time to creep failure in the SPCT closely matches that of a traditional UCT under identical temperature conditions. In situations lacking pre-existing data on the material's stress-dependent behavior, a specific ratio is used to equate the SPCT load (F) to the UC stress (σ). This relationship shown in Eq 11 applies specifically to 0.5-mm-thick specimens per the standard (Ref 24). This ratio is critical for the SPCT to accurately mimic UCT conditions and provide reliable, transferable data for various material testing needs. The standard suggested to use the following relation to design the test matrix at the initial stage and then

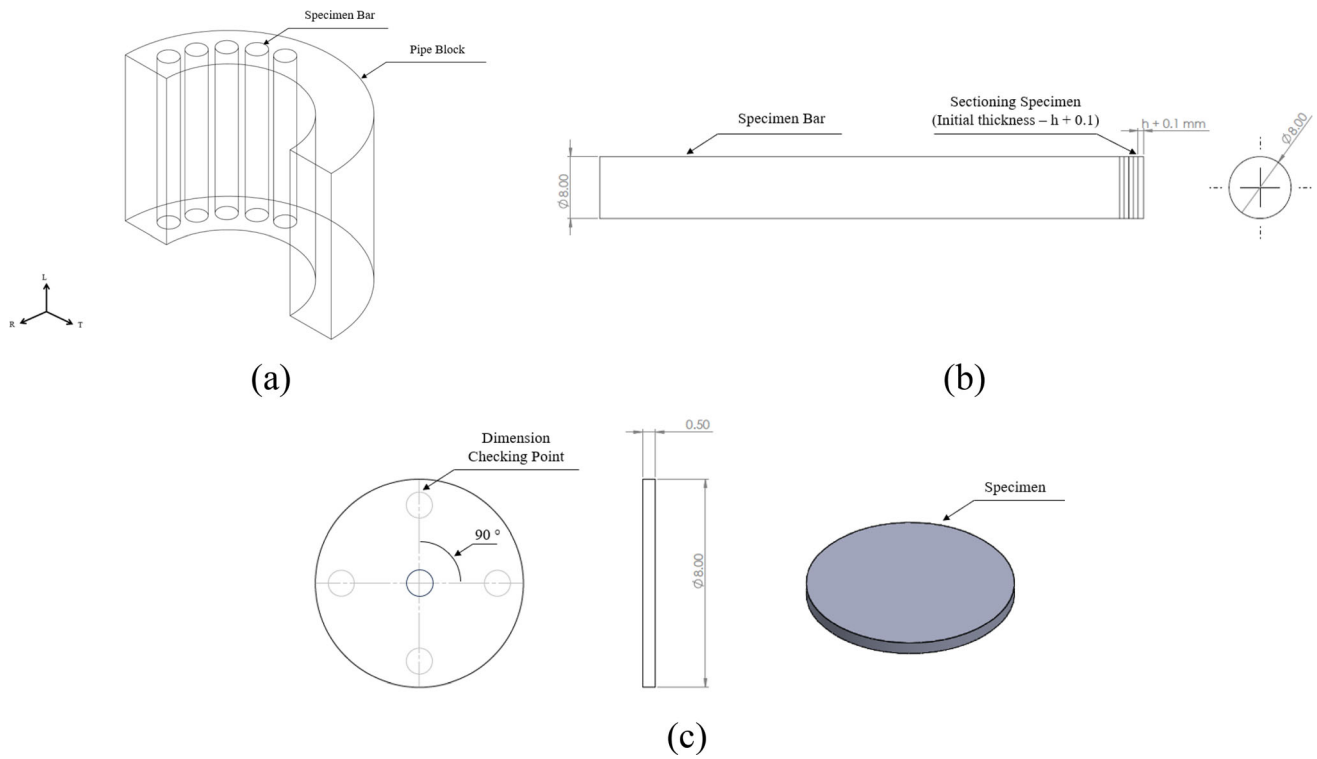


Fig. 5 (a) Outlined diagram of the extracted bar, (b) slicing into disk shaped sample, (c) prepared samples

proceed to evaluate suitable correlation factor based on the rupture life of SPCT and UCT.

$$\Psi = F/\sigma = 1.89 \quad (\text{Eq 11})$$

This section examines the relationship between rupture time (t_r) and applied stress (σ) for both standard UCT and SPCT at 600 °C. The relationship, depicted in Fig. 9, is based on Chakrabarty's membrane stretch theory combined with sophisticated regression analysis (Ref 16). An important aspect of this analysis is the adjustment of the Ψ factor from 1.89 to 2.5, which allows accurate plotting of the SPCT data together with the UCT data. The rationale behind selecting a Ψ value of 2.5 is that this scaling factor enables matching of the rupture life between the small punch test load and the equivalent uniaxial creep stress. In other words, a Ψ of 2.5 provides the conversion between SPCT load and UCT stress that equates time-to-rupture or failure life under the present test conditions. This optimized correlation factor was determined through comparative testing and analysis by incrementally adjusting Ψ until the rupture life curves aligned between the two test methods. A prominent observation in both datasets is the manifestation of a power-law dependence between rupture time and stress, concisely expressed in Eq 12. Such power-law relationships are fundamental for understanding material behavior under stress and bridging the gap between theoretical constructs and practical testing. The ability to correlate UCT and SPCT data through this relationship provides comprehensive insights into material performance under the specified conditions.

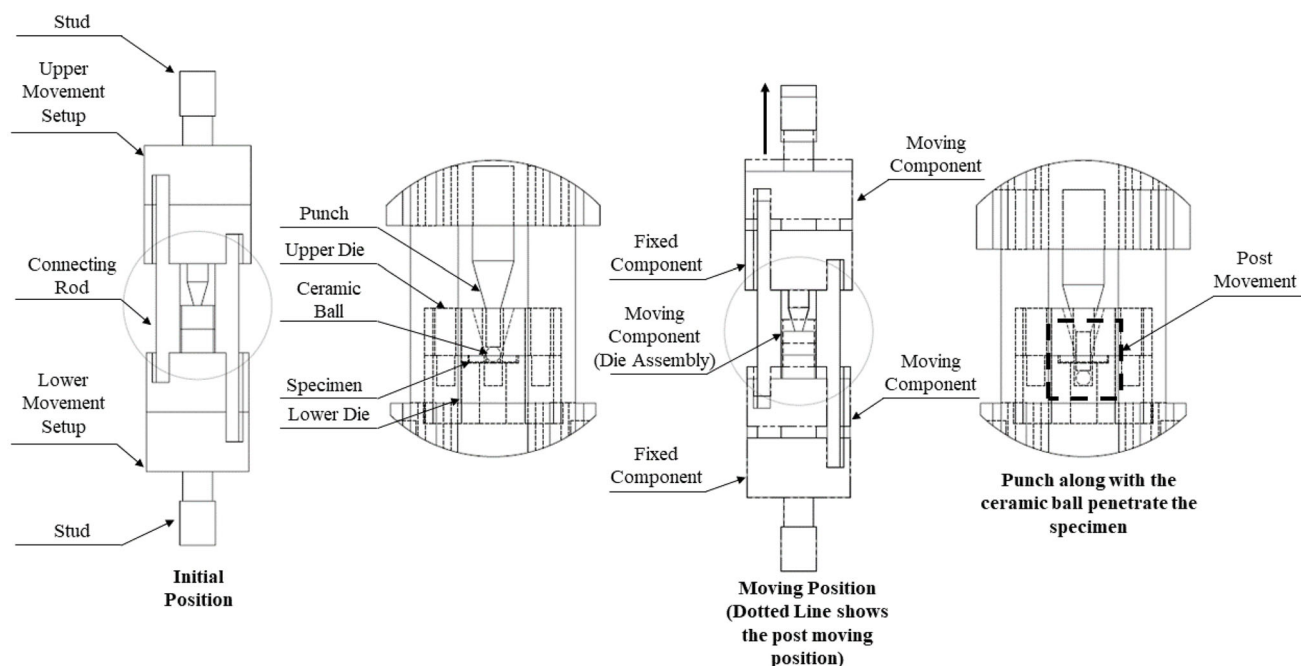
$$t_r = M\sigma^{-v} \quad (\text{Eq 12})$$

where M is defined as a material constant that varies with temperature. A thorough examination demonstrates a pronounced linear correlation between rupture time and creep

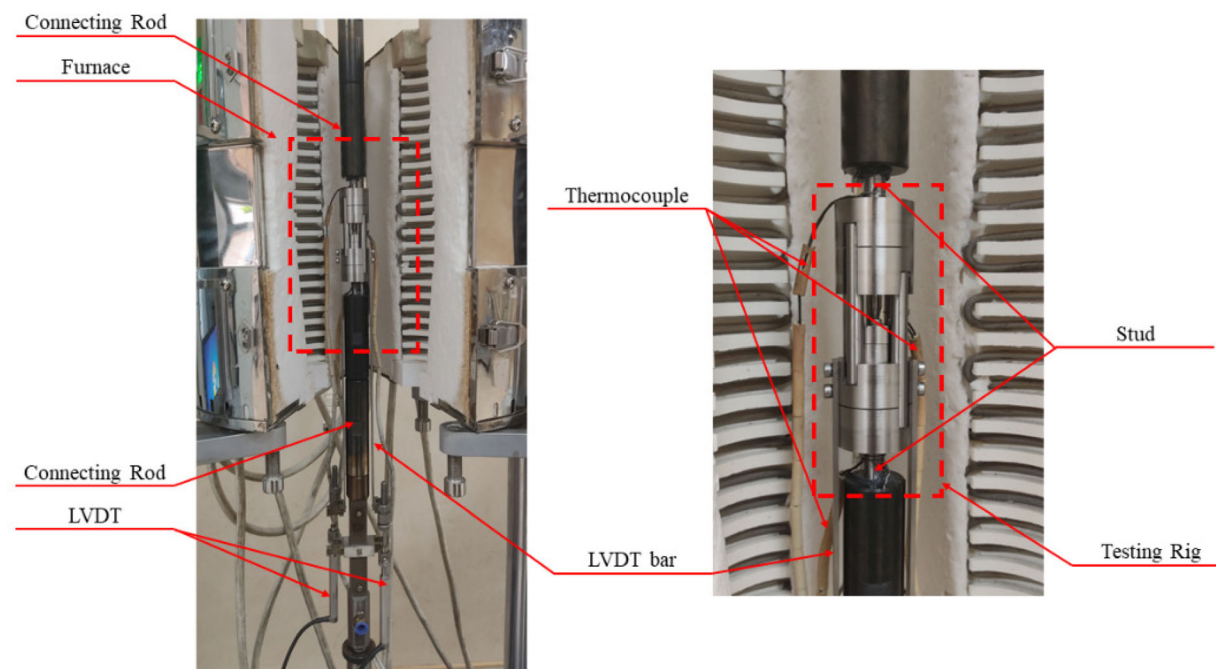
stress for both SPCT and UCT, particularly when assessed applying logarithmic scale on both axes. Interestingly, the non-negative value of v observed in the SPCT data closely aligns with the findings from the UCT at a temperature of 600 °C. When comparing the regression lines of these two sets of data, it becomes evident that employing a Ψ value of 2.5 leads to a close approximation between the rupture times observed in SPCTs and those in conventional UCTs for Grade 91 steel, particularly at 600 °C.

In SPCT, the stress applied is biaxial (Ref 26), meaning it acts in two different directions, whereas UCT involves uniaxial stress, or stress applied in a single direction. This difference in stress distribution can lead to variations in how the material deforms and ultimately fails. Biaxial stress might result in more complex stress states within the material, potentially affecting the microstructural changes differently compared to uniaxial stress. Another fact the application of compressive forces can induce different microstructural changes compared to the tensile forces in UCT. These changes might affect the creep properties and rupture time. High loading and temperature conditions in SPCT could accelerate certain microstructural transformations, leading to an overestimation of the material's lifespan. While further research is needed to fully elucidate these phenomena, such investigations are beyond the current scope. Nonetheless, the obtained data, apart from the noted discrepancies, provide ample opportunity for additional in-depth studies in this emerging field. Table 5 gives the material constant of Grade 91 steel obtained from SPCT and UCT.

An additional analysis was conducted to evaluate the applicability of the correlation factor of 2.5 for a larger dataset. Figure 9 incorporates short-term creep data from National Institute for Materials Science (NIMS) (Ref 27) for Grade 91 pipe and tube at 600 °C and stresses above 125 MPa in which the long-term creep data are excluded., The transition from



(a)



(b)

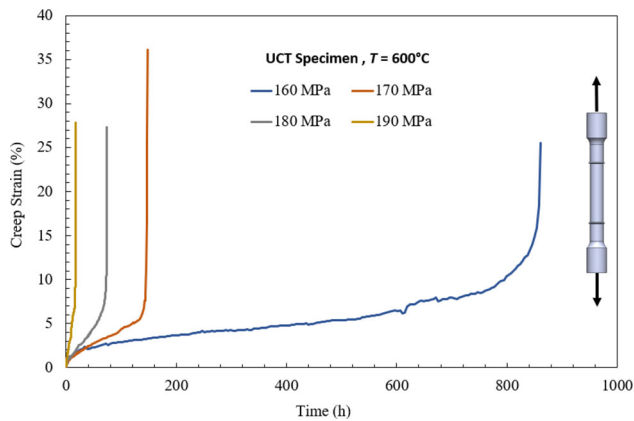
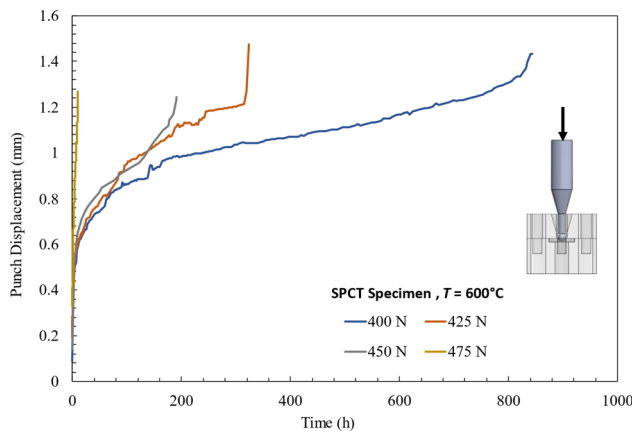
Fig. 6 (a) Details and working principle of the test rig, (b) experimental setup

short- to long-term creep occurred at 50% yield strength. For Grade 91 steel at 600 °C, the yield strength is approximately 250 MPa (Ref 28). Two lines representing factors of $2.5 \pm 10\%$ were added in Fig. 9. Clearly, all the data on rupture life lay within these two lines, confirming that a correlation factor ranging from 2.25 to 2.75 can accurately forecast rupture life across a broad spectrum of stress levels.

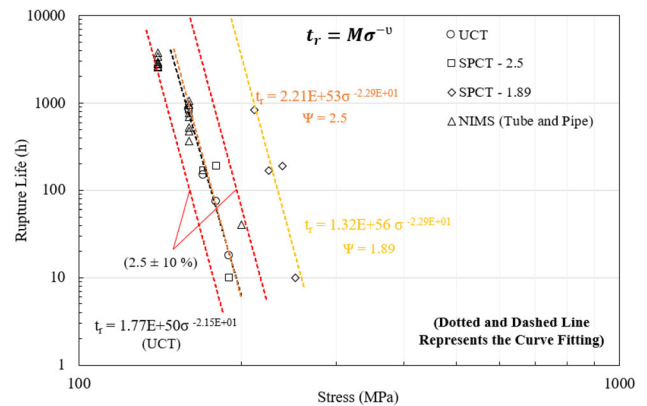
Additionally, the lower stress data show a tendency toward the $2.5 + 10\%$ line, potentially indicating a transition in the dominant creep damage mechanisms at these stresses. The trend toward higher Ψ at lower stresses warrants further investigation but overall proves this approach can extrapolate small punch results to estimate creep rupture life. Evaluating extended datasets helps validate correlation factors and pro-

Table 3 Experimental plan of the SPCT

ID-specimen	Type of the specimen	Load, N	T, °C
SPCT_400	As-received specimen	400	600 °C
SPCT_425		425	
SPCT_450		450	
SPCT_475		475	

**Fig. 7** Creep strain versus time for Grade 91 steel at 600 °C**Fig. 8** Punch displacement versus time for Grade 91 steel at 600 °C**Table 4** Uniaxial and small punch creep test data for Grade 91 steel at 600 °C under different stress level and loading condition.

Type of the specimen	T °C	Load, N	t_r , h
Small punch specimen	600 °C	400	843
		425	320
		450	191
		475	10
		Stress (MPa)	
Uniaxial smooth specimen		160	861
		170	148
		180	74
		190	18

**Fig. 9** The relationship between rupture life and creep stress (or the equivalent stress for SPC load) for both uniaxial and small punch tests across various Ψ values

vides confidence for applying small punch testing in practical component life assessment.

4.3 Creep Properties Evaluation using Monkman–Grant Creep Model

In traditional creep testing, the minimum creep rate is an essential measure for evaluating the remaining lifespan of components. On the other hand, the SPCT primarily yields information regarding the displacement of the punch over time. To enable meaningful comparisons with UCT data, these displacement measurements must be converted into equivalent strain values. In this study, two methodologies from previous works have been adopted (Ref 17, 21) to correlate deflection and strain. As expressed in Eqs 4, 5, 6 and 7, the relationship between punch displacement and strain (ϵ_{eq}) has been established. Details derivation of Eqs 4, 5, 6 and 7 is presented in Sect. 2.1. These equations act as a bridge, translating the observed punch displacements in the small punch test into quantifiable strain values. This conversion is vital for comparing and contrasting the small punch test results with uniaxial creep test findings, furnishing a comprehensive perspective of the material's creep behavior. Using these established equations, the minimum displacement rate from the small punch tests to the MCSR can be correlated. This enables comparing how the MCSR, or equivalent MCSR from small punch displacement, influences the time of rupture in both testing conditions at 600 °C. As seen in Fig. 10, both datasets exhibit a strong linear relationship when plotted on double logarithmic scales, indicating a direct link between rupture time and minimum creep rate.

This finding aligns well with the renowned Monkman–Grant relation (Ref 29) expressed in Eq 8, which holds true for the

Table 5 Material constants of Grade 91 steel at 600 °C for comparison between UCT and SPCT

Material parameters	M	<i>v</i>
SPCT	2.21×10^{53}	22.9
UCT	1.77×10^{50}	21.5

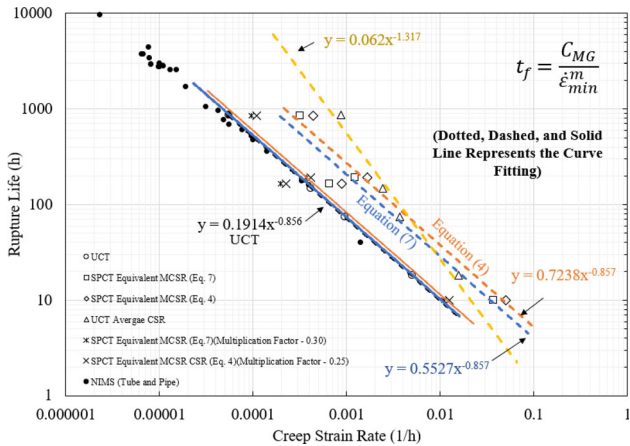


Fig. 10 The relationship between rupture time and minimum creep rate (or the adapted creep rate for SPCT deflection) in both uniaxial and small punch tests at 600 °C

Table 6 Material constants of Grade 91 steel at 600 °C for comparison between UCT and SPCT using Monkman–Grant model

Material parameters	C_{MG}	m
SPCT (Eq 7)	0.5527	0.857
SPCT (Eq 4)	0.7238	0.857
UCT (MCSR)	0.1914	0.856
UCT (ACSR)	0.2695	1.019

data from both test methods. This demonstrates the applicability of the Monkman–Grant relation across different creep testing techniques. Furthermore, Table 6 compiles detailed constants pertaining to the relation for different materials, providing comprehensive insights into creep performance. Additionally, Fig. 10 includes the average creep strain rate data from the UCT and MCSR data from NIMS (Tube and Pipe). Interestingly, the equivalent MCSR from the small punch tests agrees well with the average rate from the uniaxial tests. This can be attributed to the small sample size as well as concentrated stress state in SPT, which shortens the initial transient stage and diminishes the tertiary creep phase. The specimen likely enters and remains in steady-state creep for most of the test duration. As the brief initial and final stages contribute minimally, the average and minimum strain rates become very close. In addition, two multiplication factors were determined to match the fit lines from Eqs 4, 5, 6 and 7 to the UCT data fit line. A multiplier of 0.3 was used for Eq 7, and 0.25 was applied to Eq 4. As evident in the figure, with these adjustments, the lines from both conversion methodologies align well with the uniaxial creep test results and the data from

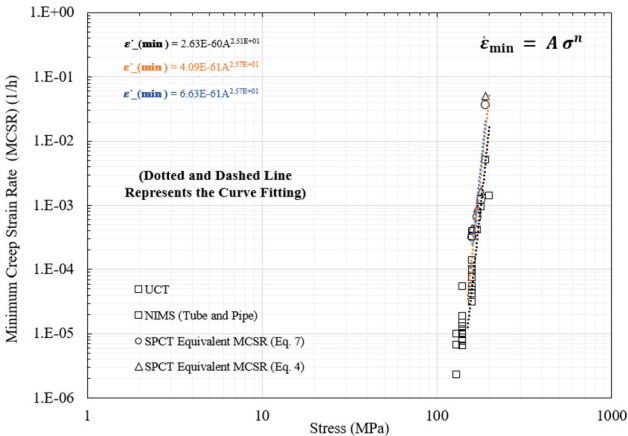


Fig. 11 The relationship between the minimum creep rate (or the adjusted strain rate for SP deflection) and creep stress (or the estimated stress for SP load) in both uniaxial and small punch tests

the NIMS. Furthermore, when these results are compared and aligned with the data from NIMS, it becomes clear that this methodology is effective for a broader dataset. This elaboration not only confirms the reliability of the approach but also underscores its potential applicability in extensive data analysis scenarios. However, further examination is still needed to utilize these magnification factors to construct full creep curves from the small punch test data for complete representation of the creep strain evolution. Additional work is required to validate the methodology across the entire creep life. Nevertheless, the strong correspondence achieved between the adjusted small punch strain rates and uniaxial rates proves these conversion factors can translate between the two techniques.

By correlating small punch displacement rates to equivalent MCSR, the well-established Monkman–Grant relation for rupture life prediction could be applied to SPCT data. The excellent overall agreement with uniaxial creep data verifies that this physics-based framework can be leveraged to extrapolate short-term small punch tests for life assessment purposes. However, the data point at 170 MPa equivalent stress for the SPCT deviates from the trendline, as noted previously. The underlying reasons for this variance were discussed in prior sections. Nevertheless, as stated before, the remainder of the data align well with the trendline and provides sufficient data for further analysis. While further validation and refinement is needed, particularly to understand deviations at certain stresses, this significantly advances the applicability of small punch creep testing and evaluating creep properties using Monkman–Grant model.

4.4 Creep Properties Evaluation using Norton Creep Model

Figure 11 illustrates the relationship between the MCSR ($\dot{\epsilon}_{min}$) and applied stress or equivalent stress for both creep tests conducted at 600 °C. As observed, the correlation between MCSR and creep stress follows the well-known power-law relationship described by the Norton Law (Ref 22), expressed in Eq 9. Figure 11 presents the MCSR versus applied stress on double logarithmic scales, revealing a clear linear correlation for both UCT and SPCT. Interestingly, a minor variation is observed in the stress exponent ' n ' values between the SPCT and UCT data. This slight difference implies that across a broad

stress range, predicting MCSR from small punch deflection data closely agrees with uniaxial data for Grade 91 steel. This suggests similarities in the underlying creep deformation mechanisms for both test methods.

Furthermore, comparing the trendlines shows that the MCSR predicted by SPCT, particularly with a Ψ value of 2.5, align closely with the uniaxial rates within the medium stress band of 160-190 MPa. This confirmation underscores the reliability of SPCT in accurately predicting the creep behavior of Grade 91 steel within a defined stress range, reinforcing its compatibility with the Norton creep model. Additionally, the inclusion of data from the NIMS within Fig. 11 further corroborates the findings of this research, demonstrating a strong alignment between the external data and the outcomes of this study. In addition, Table 7 compiles detailed constants pertaining to the relation for different materials, providing comprehensive insights into creep performance.

4.5 Creep Properties Evaluation using Larson–Miller Parameter

Over the years, extrapolating short-term creep data to predict long-term rupture life of materials has been an important research focus. A key technique used for this extrapolation involves applying a time-temperature parameter (TTP). Many different TTPs have evolved to analyze the complex relationship between creep-rupture time and test temperature in metals. Well-known examples include the Manson-Haferd parameter, Fisher-Dorn parameter, Sub-Aviation parameter, Orr-Sherby-Dorn parameter, and the Larson–Miller parameter (LMP) (Ref 30, 31).

A significant advantage of these parametric techniques is that they require only a limited amount of data to construct a detailed curve (Ref 32). The LMP is a time-temperature parameter that can be used to extrapolate short-term creep data to predict long-term rupture life in metallic materials (Ref 33). The LMP approach accounts for the linked influences of time and temperature, enabling the correlation of creep rupture time from various temperatures into a unified curve. On this curve, the logarithm of stress is plotted against the Larson–Miller parameter, which can subsequently be employed to accurately forecast creep rupture data. The original Larson–Miller model is based on the Arrhenius equation and evaluates the influence of stress and temperature on creep life. It posits that the activation energy varies as a function of stress (Ref 4).

This study utilizes the LMP approach to analyze the efficacy of the relationship shown in Eq 10 for Grade 91 steel and help identify the suitable correlation factor Ψ value. For materials like Grade 91 steel, the value of C is generally suggested to be 33 (Ref 34). Importantly, this method does not absolutely employ failure time to assess properties. Rather, it concurrently assesses the temperature effects on the rupture time by

Table 7 Material constants of Grade 91 steel at 600 °C for comparison between UCT and SPCT using Norton creep model

Material parameters	A	n
SPCT (Eq 7)	4.09×10^{-60}	25.1
SPCT (Eq 4)	6.63×10^{-61}	25.7
UCT	2.63×10^{-60}	25.7

converting the rupture time into a Larson–Miller parameter number. For each load level, LMPs were determined for small punch, uniaxial tests and NIMS data and graphed against stress in Fig. 12. The LMPs increased linearly as creep stress decreased in both test types, indicating that the LMP method can be applied to small punch creep data. At an equivalent stress of 180 MPa, the small punch LMP overestimated the trendline fit to the data. This overestimation could be attributed to microstructural changes or localized deformation mechanisms specific to small punch testing at this stress level. Aside from this anomaly, the remaining data align well with the uniaxial test results and are adequate for further analytical purposes. Overall, good agreement was found between small punch and uniaxial data using a correlation factor Ψ of 2.5.

The LMP values ranged from approximately 30216 to 32296 for stresses between 160 and 190 MPa. The relationship between stress (σ) and LMP can be expressed as $\sigma = a \times \text{LMP} + b$, where a and b are constants determined by linear regression. For the small punch creep test data, $a \approx -0.0169$ and $b \approx 692.59$, while for the uniaxial creep test data, $a \approx -0.0209$ and $b \approx 815.38$. This analysis demonstrated that the LMP method can be used with small punch creep tests on Grade 91 steel to extrapolate short-term data and predict long-term rupture life. A correlation factor of 2.5 provided good agreement between small punch and uniaxial creep rupture data. The LMP method is a useful parametric tool for evaluating and predicting creep behavior.

4.6 Fractographic Analysis

4.6.1 As-Received Material. Figure 13 showcases the distinctive microstructure of Grade 91 steel. Prominently visible in the micrograph are elongated needle-like grains known as laths, which are characteristic of a martensitic microstructure. This lath structure is formed when the steel undergoes rapid cooling from a high-temperature austenitic phase, resulting in the transformation to martensite (Ref 35). Additionally, the micrograph reveals bright, straight lines separating the laths, which are known as prior austenite grain boundaries (PAGBs). These boundaries represent the original grain structure of the steel before it underwent the transformation to martensite during the cooling process (Ref 36). Scattered throughout the microstructure are numerous small, white dots,

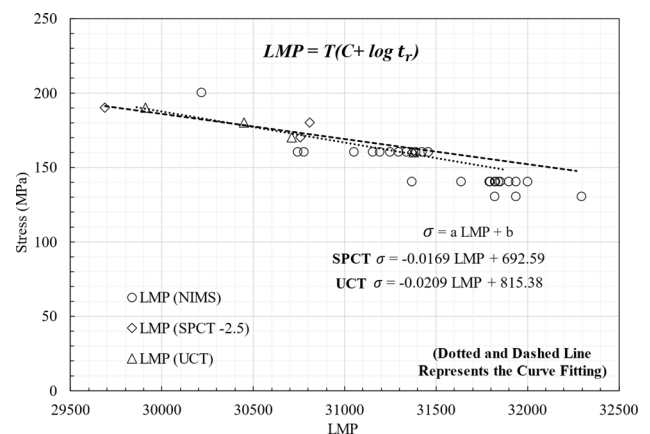


Fig. 12 Dependence of stress on LMP for Grade 91 Steel at 600 °C of UCT and SPCT

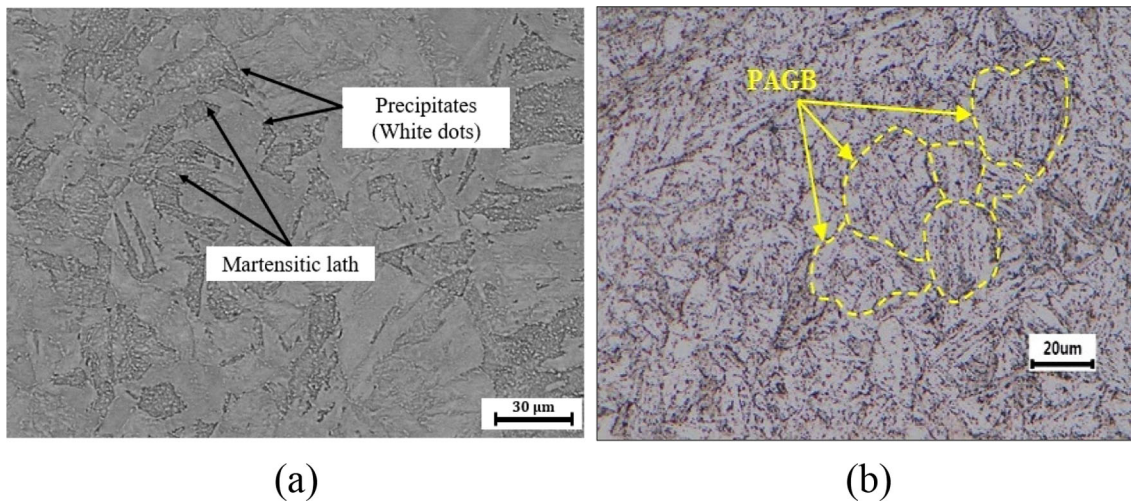


Fig. 13 Micrograph of as-received Grade 91: (a) SEM, (b) optical

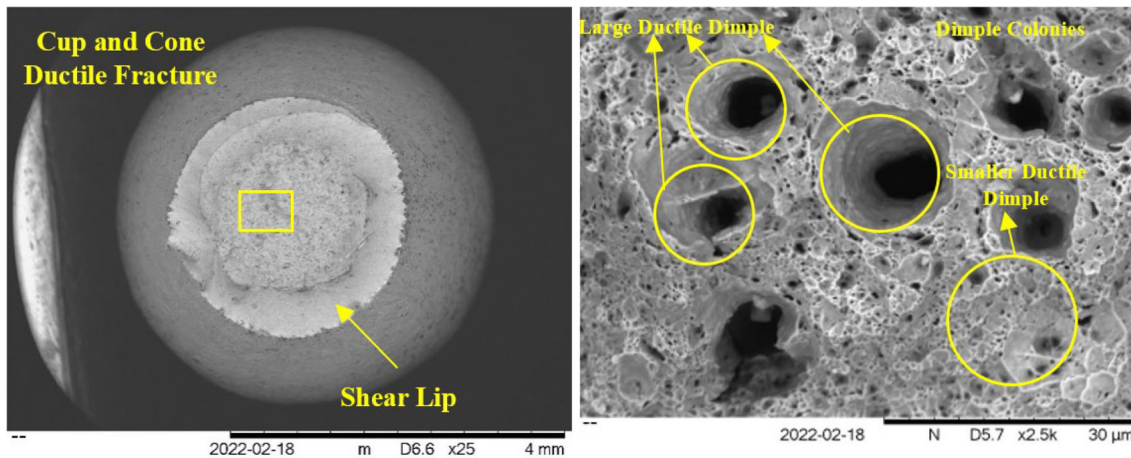


Fig. 14 Presentation of the fracture surface on the smooth specimen using scanning electron microscopy, at a stress level of 180 MPa and a temperature of 600 °C

which are precipitates. These precipitates are fine particles of various compounds that have formed from the steel's chemical composition during the cooling and heat treatment processes. In Grade 91 steel, typical precipitates include $M_{23}C_6$ carbides and MX carbonitrides (Ref 36). The presence of the tempered martensitic lath structure, combined with the fine distribution of precipitates, contributes to the exceptional properties of Grade 91 steel. The interconnected network of lath grains provides high strength and resistance to deformation, while the fine precipitates act as obstacles to the movement of dislocations, thereby enhancing the material's resistance to creep deformation at elevated temperatures (Ref 35, 36). Furthermore, the tempered martensite microstructure offers a favorable balance between strength and toughness, ensuring robust performance under various loading conditions.

4.6.2 Crept Specimen. Figure 14 displays a SEM image showing the fracture surface of a smooth specimen that underwent testing at 180 MPa. The micrograph exhibits the characteristic cup-cone fracture morphology and shear lip that are commonly observed. Further analysis of the fracture surface

revealed uniformly distributed dimples of differing sizes, indicative of ductile fracture. The formation and coalescence of micro-voids along grain boundaries lead to the generation of these ductile dimples. The presence of relatively large and deep dimples suggests high ductility and enhanced creep life of the material (Ref 37).

Figure 15 shows the crept specimen under SPCT at different loading conditions. At lower load, a distinct separation of the cap from the main specimen is evident, indicating a localized deformation phenomenon. This detachment suggests that the applied stress levels were insufficient to induce significant deformation throughout the entire specimen volume, leading to a preferential deformation concentrated within the cap region. In contrast, under high loading conditions, the cap appears to remain firmly attached to the main specimen, implying a more uniform deformation behavior across the entire volume. Additionally, the specimen shows similar types of deformation for all loading conditions. Figures 16 and 17 illustrate the macroscopic morphology of the fracture surface for a specimen tested at 400N and 475N at 600 °C, respectively. The load levels were chosen to determine the fracture characteristics at

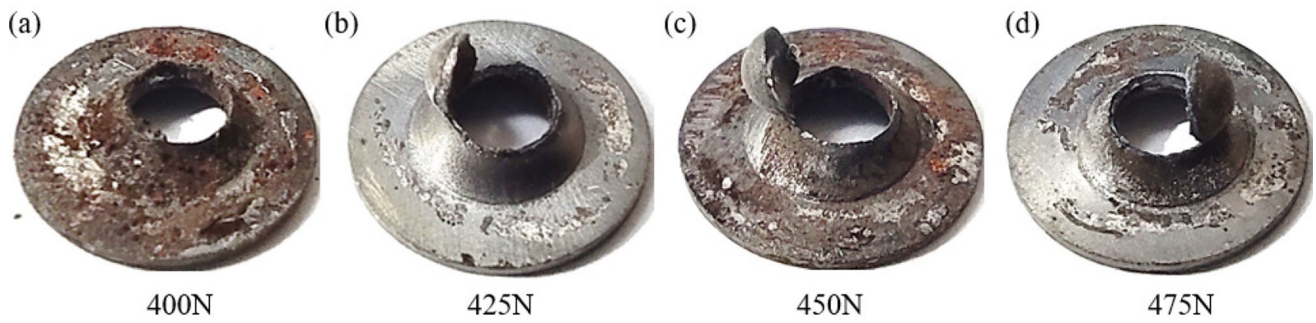


Fig. 15 Small punch crept specimen: (a) 400N, (b) 425N, (c) 450N, (d) 475N

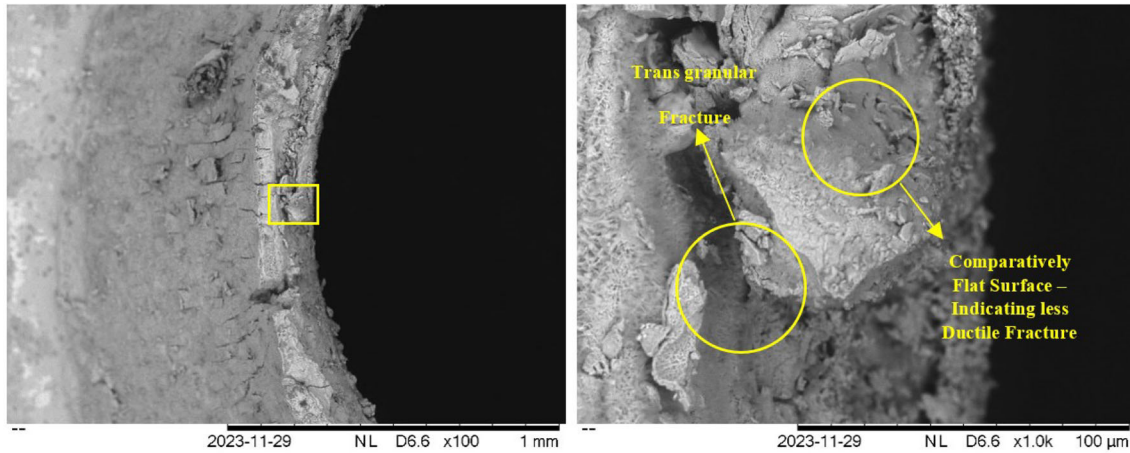


Fig. 16 Presentation via scanning electron microscopy of the fracture surface on the small punch specimen under a 400N load at 600 °C

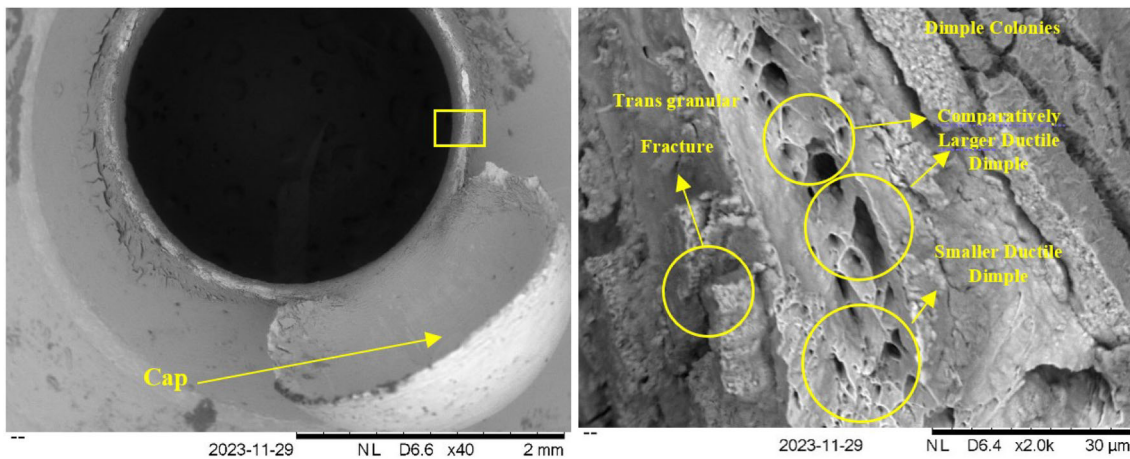


Fig. 17 Presentation via scanning electron microscopy of the fracture surface on the small punch specimen under a 475N load at 600 °C

lower and higher values. The post-test shape displays the fracture morphology with a central circular hole and indented structure on the fracture surfaces, consistent across all load levels. The loading condition in small punch creep testing can be considered as having both perpendicular and parallel force components relative to the specimen surface due to the spherical indenter (Ref 38). The perpendicular forces contribute

to thickness reduction while the parallel forces promote circumferential crack propagation, resulting in the observed surface indentations. Fractographic analysis reveals similar ductile dimple-dominated creep fracture behavior between uniaxial and small punch creep tests (Fig. 16 and 17). Figure 16 shows that comparatively less ductile fracture dominates the rupture, confirming that at lower load levels the fracture is

brittle in nature (Ref 39). Narrow transgranular crack regions are also evident near the fracture center for the 475N load level (see Fig. 17). The transition from predominantly intergranular fracture initially to mixed inter/transgranular cracks at intermediate times and finally transgranular fracture near rupture indicates a mixed fracture mode. Based on these findings, it can be concluded that at lower load levels the brittle fracture dominates, and as the load level increases, the fracture mechanism tends to shift to ductile fracture. Previous studies have correlated dimple formation with ductile fracture mechanisms under high tensile stresses (Ref 40, 41).

5. Conclusion

This research demonstrates the effectiveness of SPCT in assessing the creep performance and rupture lifespan of Grade 91 steel, compared to traditional UCT. Key findings include:

- SPCT creep strain evolution matched UCT behavior, exhibiting typical creep regimes.
- A correlation factor of 2.5 provided excellent agreement between SPCT and UCT rupture times at 600 °C.
- Established rupture life prediction models (Monkman–Grant, Larson–Miller) were successfully applied to SPCT data.
- Minimum creep rate versus stress trends aligned closely between test types when applying the correlation factor.
- Fractographic analysis revealed similar ductile, dimple-dominated fracture morphologies in both test types.

SPCT, with appropriate correlation, reliably reproduces creep performance evaluated from conventional UCT for Grade 91 steel. This enables accelerated creep screening using minimal material volumes. Further validation across wider test conditions is recommended.

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